

Concept 3 | Properties and Graphs of Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Amplitude

$$y = a \sin(bx + c) + d \Rightarrow |a|$$

Period

$$T = \frac{2\pi}{|b|}$$

Range

$$d - |a| \leq y \leq d + |a|$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q001 Written

(1) Express $\sin \frac{9\pi}{5}$ as a trigonometric function of an acute angle. (2) Simplify $\sin \theta + \sin(\theta + \frac{\pi}{2}) + \sin(\theta + \pi) + \sin(\theta + \frac{3\pi}{2})$.

Solution

1. Reduce the angle to its reference angle and keep the quadrant sign.
2. Use the quarter-turn identities: $\sin(\theta + \frac{\pi}{2}) = \cos \theta$, $\sin(\theta + \pi) = -\sin \theta$, and $\sin(\theta + \frac{3\pi}{2}) = -\cos \theta$.

Final answer

(1) $-\sin \frac{\pi}{5}$ (2) 0

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q002 Written

(1) Express $\cos(-\frac{\pi}{6})$ as a trigonometric function of an acute angle. (2) Express $\sin \frac{8\pi}{7}$ as a trigonometric function of an acute angle. (3) Simplify $\cos(\frac{\pi}{2} - \theta) \sin(\pi - \theta) - \sin(\frac{3\pi}{2} + \theta) \cos(\pi - \theta)$.

Solution

1. Use evenness of cosine and the reference-angle rules.
2. Replace shifted functions by $\sin \theta$ or $\cos \theta$, then simplify.

Final answer

(1) $\cos \frac{\pi}{6}$ (2) $-\sin \frac{\pi}{7}$ (3) $\sin^2 \theta - \cos^2 \theta$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q003 Written

Solve all three given demands: (1) express $\cos \frac{17\pi}{5}$, $\cos \frac{5\pi}{7}$, $\sin(-\frac{7\pi}{8})$, $\sin(-\frac{19\pi}{8})$ using acute angles; (2) find $\tan(-\frac{\pi}{6})$ and $\tan(-\frac{\pi}{4})$; (3) simplify (a) $\sin \theta + \cos(\pi + \theta) - \cos(\pi - \theta) + \sin(\pi + \theta)$, (b) $\sin(\theta + \frac{\pi}{2}) \cos(\theta + \pi) + \sin(-\theta) \cos(\frac{\pi}{2} - \theta)$.

Solution

1. Reduce each angle to an acute reference angle and apply the sign in its quadrant.
2. For (3), use the quarter-turn and half-turn identities before collecting terms.

Final answer

$$(1) -\cos \frac{2\pi}{5}, -\cos \frac{2\pi}{7}, -\sin \frac{\pi}{8}, -\sin \frac{3\pi}{8} \quad (2) -\frac{1}{\sqrt{3}}, -1 \quad (3a) 0, (3b) -1$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q004 Written

Simplify $\sin(\frac{3\pi}{2} - \theta) \sec(\pi + \theta) - \tan(\theta - \frac{\pi}{2}) \cot(\frac{5\pi}{2} + \theta)$.**Solution**

1. Use $\sin(\frac{3\pi}{2} - \theta) = -\cos \theta$, $\sec(\pi + \theta) = -\sec \theta$, $\tan(\theta - \frac{\pi}{2}) = -\cot \theta$, and $\cot(\frac{5\pi}{2} + \theta) = -\tan \theta$.
2. Both products equal 1, so their difference is 0.

Final answer

0

Q005 Written

Draw the parent graphs $y = \sin \theta$, $y = \cos \theta$, **and** $y = \tan \theta$, **and state each period and range.****Solution**

1. Mark one standard cycle, then repeat it periodically.
2. For tangent, draw vertical asymptotes at $\theta = \frac{\pi}{2} + k\pi$.

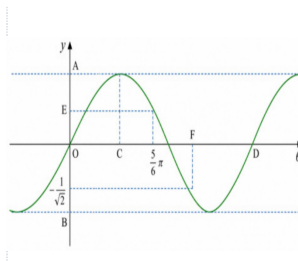
Final answer $\sin, \cos : T = 2\pi, -1 \leq y \leq 1; \quad \tan : T = \pi, y \in \mathbb{R}$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q006 Written

For the given graph of $y = \sin \theta$, determine the labelled scales A to F .



Solution

1. Read the extrema and the x-axis scale from the standard sine cycle.
2. Use $\sin \frac{5\pi}{6} = \frac{1}{2}$ and the descending branch to locate F.

Final answer

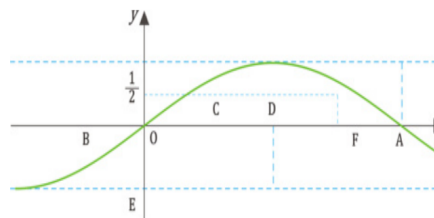
$$A = 1, B = -1, C = \frac{\pi}{2}, D = 2\pi, E = \frac{1}{2}, F = \frac{5\pi}{4}$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q007 Written

(1) Draw the parent graphs $y = \sin \theta$, $y = \cos \theta$, $y = \tan \theta$, and state their periods. (2) For the given graph of $y = \cos \theta$, determine the labelled scales A to F .



Solution

1. Use the standard parent graph and mark quarter-period points.
2. The vertical values are the cosine range $[-1, 1]$; the horizontal values follow the 2π cycle.

Final answer

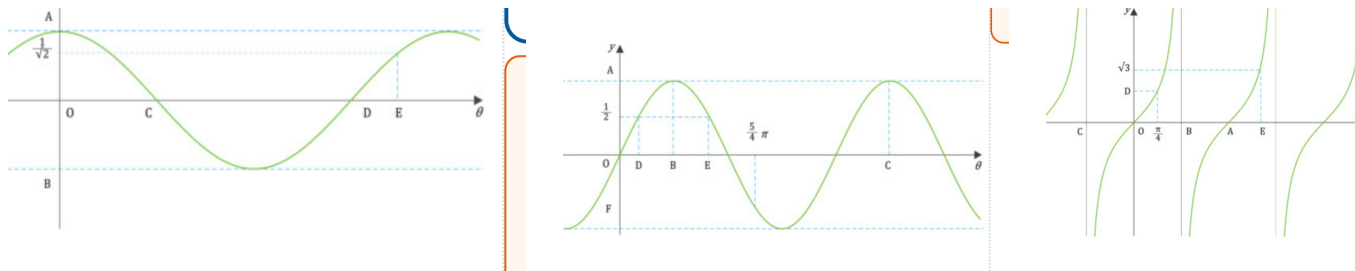
(1) $T_{\sin} = 2\pi$, $T_{\cos} = 2\pi$, $T_{\tan} = \pi$; (2) read the marked extrema, intercepts and quarter-period points from the graph

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q008 Written

Complete all four given graph tasks: draw the three parent graphs and find their periods; read the labelled scales on the given cosine graph; read the labelled scales on the given sine graph; and read the labelled scales on the given tangent graph.



Solution

1. Draw axes and the standard cycle first.
2. Read each label from an intercept, extremum, asymptote or quarter-period tick; do not change the graph scale.

Final answer

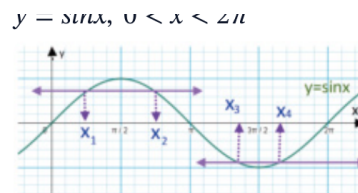
Use the standard cycles: sine/cosine have period 2π and range $[-1, 1]$; tangent has period π , range $\mathbb{R}$, and asymptotes at odd multiples of $\frac{\pi}{2}$. The requested A+F labels are read from the marked ticks and extrema in each supplied graph.

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q009 Written

The given graph represents $y = \sin x$ for $0 < x < 2\pi$. Find $x_1 + x_2 + x_3 + x_4$ from the four marked x-coordinates.



Then find: $x_1 + x_2 + x_3 + x_4$

Solution

1. Identify the y-level associated with each marked point and use the sine cycle on $(0, 2\pi)$.
2. Add the four x-coordinates only after they have been read from the same horizontal scale.

Final answer

Read x_1, x_2, x_3, x_4 from the supplied graph and add the four marked coordinates; the result is the graph-reading total.

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q010 Written

Draw the graphs of $y = 2 \cos \theta$ and $y = \tan \frac{\theta}{2}$, and find their periods.

Solution

1. The coefficient 2 outside cosine doubles the y-values but not the period.
2. For $\tan(\theta/2)$, multiply the parent tangent period by the reciprocal of $1/2$.

Final answer

$$y = 2 \cos \theta : T = 2\pi, -2 \leq y \leq 2; \quad y = \tan \frac{\theta}{2} : T = 2\pi$$

Q011 Written

Draw the graphs of $y = 2 \sin \theta$, $y = \tan 2\theta$, and $y = \cos \frac{\theta}{2}$, and find their periods.

Solution

1. Use the vertical factor for amplitude/range and the reciprocal of the coefficient of θ for horizontal scale.
2. For tangent, the parent period is π ; for sine/cosine it is 2π .

Final answer

$$T = 2\pi, \frac{\pi}{2}, 4\pi \text{ respectively; ranges } [-2, 2], \mathbb{R}, [-1, 1]$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q012 Written

Draw the nine given graphs and find each period: (a) $3 \sin \theta$, (b) $3 \cos \theta$, (c) $\frac{1}{2} \cos \theta$, (d) $\sin 3\theta$, (e) $\cos 2\theta$, (f) $\tan 3\theta$, (g) $\sin \frac{\theta}{2}$, (h) $\cos \frac{\theta}{3}$, (i) $\tan \frac{\theta}{3}$.

Solution

1. Read the vertical coefficient as the amplitude/range factor.
2. Divide the parent period by the absolute coefficient of θ .

Final answer

(a) 2π (b) 2π (c) 2π (d) $\frac{2\pi}{3}$ (e) π (f) $\frac{\pi}{3}$ (g) 4π (h) 6π (i) 3π ; ranges follow the outside coefficient

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q013 Written

Draw $y = -2|\sin(3x)|$ and find its period.**Solution**

1. The absolute value makes the sine pattern repeat after π in its input, so the x-period is $\pi/3$.
2. The negative outside factor reflects the nonnegative $|\sin(3x)|$ graph below the x-axis.

Final answer

$$T = \frac{\pi}{3}, \quad 0 \geq y \geq -2$$

Q014 Written

Draw $y = \sin(\theta - \frac{\pi}{6})$ and find its period.**Solution**

1. Start the standard sine cycle at $\theta = \pi/6$.
2. A horizontal shift does not change the parent period.

Final answer

$$\text{shift right by } \frac{\pi}{6}, \quad T = 2\pi$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q015 **Written**

Draw the graphs and find their periods: (1) $y = \sin(\theta - \frac{\pi}{3})$, (2) $y = \cos(\theta + \frac{\pi}{4})$, (3) $y = \tan(\theta - \frac{\pi}{2})$.

Solution

1. Read $\theta - p$ as a shift right by p and $\theta + p$ as a shift left by p .
2. Keep the parent period unchanged.

Final answer

Shifts: right $\frac{\pi}{3}$, left $\frac{\pi}{4}$, right $\frac{\pi}{2}$; periods $2\pi, 2\pi, \pi$.

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q016 Written

Draw all given graphs and find their periods: sine group $y = \sin(\theta + \frac{\pi}{4}), \sin(\theta - \frac{\pi}{2}), \sin(\theta + \frac{\pi}{3})$; **cosine group** $y = \cos(\theta - \frac{\pi}{3}), \cos(\theta - \frac{\pi}{4}), \cos(\theta + \frac{\pi}{6})$; **tangent group** $y = \tan(\theta + \frac{\pi}{6}), \tan(\theta - \frac{\pi}{3}), \tan(\theta - \frac{\pi}{4})$.

Solution

1. Translate the parent graph horizontally by the signed amount inside the bracket.
2. Use the same vertical range and asymptote spacing as the parent function.

Final answer

All sine/cosine periods are 2π ; all tangent periods are π . The shifts are read directly from the brackets.

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q017 Written

Draw $y = 3 \cos |x + \frac{\pi}{3}|$ and find its period.

Solution

1. Use $\cos |u| = \cos(u)$, then read the horizontal shift and vertical scale.
2. The period of cosine remains 2π .

Final answer

$$y = 3 \cos(x + \frac{\pi}{3}) \text{ since cosine is even; } T = 2\pi$$

Q018 Written

Draw the graphs and find their periods: (1) $y = 2 \cos(2\theta - \frac{\pi}{2})$, (2) $y = \tan(\frac{\theta}{2} - \frac{\pi}{6})$.

Solution

1. Factor the coefficient of θ from the bracket before reading the shift.
2. Apply shift, horizontal scale, then vertical scale.

Final answer

$$(1) y = 2 \cos 2(\theta - \frac{\pi}{4}), T = \pi, -2 \leq y \leq 2; \quad (2) y = \tan \frac{1}{2}(\theta - \frac{\pi}{3}), T = 2\pi$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q019 Written

Draw the graphs and find their periods: (a) $y = 3 \sin \frac{\theta}{2}$, (b) $y = \tan(2\theta - \frac{\pi}{2})$, (c) $y = 2 \cos(2\theta - \frac{\pi}{3})$.

Solution

1. Identify the coefficient b, factor the angle, then read the shift q/b.
2. Use the parent period divided by |b| and the outside amplitude |a|.

Final answer

$$(a) T = 4\pi, -3 \leq y \leq 3; \quad (b) T = \frac{\pi}{2}; \quad (c) T = \pi, -2 \leq y \leq 2$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q020 Written

Draw all nine given graphs and find their periods: (a) $2 \cos \frac{\theta}{2}$, (b) $3 \sin 2\theta$, (c) $2 \cos 2\theta$, (d) $2 \cos(\theta - \frac{\pi}{4})$, (e) $2 \sin(\theta - \frac{\pi}{3})$, (f) $\tan(2\theta - \frac{\pi}{3})$, (g) $3 \cos(2\theta - \frac{\pi}{2})$, (h) $3 \sin(2\theta - \frac{2\pi}{3})$, (i) $2 \cos(\frac{\theta}{2} - \frac{\pi}{4})$.

Solution

1. Rewrite every angle as $b(\theta - q/b)$.
2. Apply the transformations from the inside out and record period, range and shift.

Final answer

Periods: (a) 4π ; (b) π ; (c) π ; (d) 2π ; (e) 2π ; (f) $\frac{\pi}{2}$; (g) π ; (h) π ; (i) 4π . Ranges are set by the outside factors.

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q021 Written

Draw $y = 3 \sin |2x - \pi|$ and find its period.**Solution**

1. The absolute value makes the sine pattern repeat after π in its input; with coefficient 2 the x-period is $\pi/2$.
2. The graph is nonnegative with maximum 3.

Final answer

$$y = 3|\sin(2x - \pi)|, \quad T = \frac{\pi}{2}, \quad 0 \leq y \leq 3$$

Q022 MCQ

 $\sin(\theta + \pi)$ equals:**Solution**

1. A half-turn changes the sign of sine.

Correct option: B**Final answer**

$$-\sin \theta$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q023 MCQ

 $\cos(\theta + \pi)$ equals:**Solution**

1. A half-turn changes the sign of cosine.

Correct option: B

Final answer

$$-\cos \theta$$

Q024 MCQ

 $\sin\left(\frac{\pi}{2} - \theta\right)$ equals:**Solution**

1. Use the complementary-angle identity.

Correct option: C

Final answer

$$\cos \theta$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q025 MCQ

 $\cos(-\theta)$ equals:**Solution**

1. Cosine is even.

Correct option: A

Final answer

$$\cos \theta$$

Q026 MCQ

 $\sin(-\theta)$ equals:**Solution**

1. Sine is odd.

Correct option: B

Final answer

$$-\sin \theta$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q027 MCQ

 $\cos(\pi - \theta)$ equals:**Solution**

1. Use the supplementary-angle identity.

Correct option: B

Final answer

$$-\cos \theta$$

Q028 MCQ

 $\sin(\frac{3\pi}{2} + \theta)$ equals:**Solution**

1. Shift by three quarter-turns.

Correct option: D

Final answer

$$-\cos \theta$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q029 MCQ

 $\tan(\theta - \frac{\pi}{2})$ equals:**Solution**

1. Use the quarter-turn tangent identity.

Correct option: D

Final answer $-\cot \theta$

Q030 MCQ

The period of $y = \sin \theta$ is:**Solution**

1. One sine cycle spans 2π .

Correct option: C

Final answer 2π

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q031 MCQ

The range of $y = \cos \theta$ is:**Solution**

1. Cosine is bounded by -1 and 1.

Correct option: B

Final answer

$$-1 \leq y \leq 1$$

Q032 MCQ

The period of $y = \tan \theta$ is:**Solution**1. Tangent repeats every π .

Correct option: B

Final answer

$$\pi$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q033 MCQ

A vertical asymptote of $y = \tan \theta$ occurs at:**Solution**

1. Cosine is zero at odd multiples of $\pi/2$.

Correct option: B**Final answer**

$$\theta = \frac{\pi}{2} + k\pi$$

Q034 MCQ

The maximum value of $\sin \theta$ is:**Solution**

1. Read the top of the standard sine graph.

Correct option: C**Final answer**

1

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q035 MCQ

The x-coordinate of the first positive sine maximum is:**Solution**

1. Sine first reaches 1 at $\pi/2$.

Correct option: B**Final answer** $\pi/2$

Q036 MCQ

At $\theta = 0$, $\cos \theta$ equals:**Solution**

1. The cosine graph starts at its maximum.

Correct option: C**Final answer**

1

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q037 MCQ

At $\theta = 0$, $\tan \theta$ equals:**Solution**

1. Tangent passes through the origin.

Correct option: B**Final answer**

0

Q038 MCQ

The range of $y = 3 \sin \theta$ is:**Solution**

1. Multiply the sine range by 3.

Correct option: B**Final answer** $-3 \leq y \leq 3$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q039 MCQ

The period of $y = \sin 4\theta$ is:**Solution**1. Use $2\pi/|b|$.**Correct option:** B**Final answer** $\pi/2$

Q040 MCQ

The period of $y = \cos(\theta/3)$ is:**Solution**1. The coefficient is $1/3$, so multiply the parent period by 3.**Correct option:** D**Final answer** 6π

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q041 MCQ

The period of $y = \tan 5\theta$ is:**Solution**1. Use $\pi/|b|$.**Correct option:** A**Final answer**

$$\pi/5$$

Q042 MCQ

The range of $y = \frac{1}{2} \cos \theta$ is:**Solution**

1. The outside factor halves the cosine values.

Correct option: C**Final answer**

$$-\frac{1}{2} \leq y \leq \frac{1}{2}$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q043 MCQ

The graph of $y = -2 \sin \theta$ is the parent sine graph:**Solution**

1. The negative sign reflects; 2 stretches.

Correct option: B**Final answer**

reflected and stretched vertically

Q044 MCQ

The period of $y = \tan(\theta/2)$ is:**Solution**1. Multiply the tangent period by the reciprocal of $1/2$.**Correct option:** C**Final answer** 2π

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q045 MCQ

The period of $y = \sin(3\theta)$ is:**Solution**

1. Use $2\pi/3$.

Correct option: B**Final answer** $2\pi/3$

Q046 MCQ

The graph of $y = \sin(\theta - \pi/4)$ is shifted:**Solution**

1. $\theta - p$ means shift right by p .

Correct option: B**Final answer**right by $\pi/4$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q047 MCQ

The graph of $y = \cos(\theta + \pi/3)$ is shifted:**Solution**1. $\theta + p$ means shift left by p .

Correct option: B

Final answerleft by $\pi/3$

Q048 MCQ

A horizontal shift changes the sine period from 2π to:**Solution**

1. Translation does not change period.

Correct option: B

Final answer 2π

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q049 MCQ

A horizontal shift changes the tangent period from π to:**Solution**

1. Translation does not change period.

Correct option: B**Final answer** π

Q050 MCQ

The graph of $y = \sin(\theta + \pi/2)$ equals:**Solution**

1. Use the quarter-turn identity.

Correct option: C**Final answer** $\cos \theta$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q051 MCQ

The graph of $y = \cos(\theta - \pi)$ equals:**Solution**1. A π shift reverses cosine.

Correct option: B

Final answer

$$-\cos \theta$$

Q052 MCQ

The graph of $y = \tan(\theta - \pi/2)$ has its central zero at:**Solution**1. Set $\theta - \pi/2 = 0$.

Correct option: B

Final answer

$$\theta = \pi/2$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q053 MCQ

Since cosine is even, $\cos |x|$ equals:**Solution**

1. Evenness removes the absolute value inside cosine.

Correct option: A**Final answer** $\cos x$

Q054 MCQ

In $y = a \sin(b\theta - q)$, the horizontal shift is:**Solution**

1. Factor b from the angle first.

Correct option: B**Final answer** q/b

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q055 MCQ

The period of $y = 2 \cos(3\theta - \pi)$ is:**Solution**

1. The coefficient of θ is 3.

Correct option: B**Final answer** $2\pi/3$

Q056 MCQ

The range of $y = 4 \sin(2\theta - 1)$ is:**Solution**

1. The outside factor controls the range.

Correct option: C**Final answer** $[-4, 4]$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q057 MCQ

For $y = \tan(2\theta - \pi/3)$, the shift is:**Solution**

1. Rewrite $2(\theta - \pi/6)$.

Correct option: B**Final answer**right by $\pi/6$

Q058 MCQ

The period of $y = 3 \sin(\theta/2)$ is:**Solution**

1. The coefficient is $1/2$.

Correct option: D**Final answer** 4π

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q059 MCQ

The correct transformation order for $y = af(b\theta - q)$ starts with:**Solution**

1. Factor b and read q/b first.

Correct option: B**Final answer**

horizontal shift

Q060 MCQ

The period of $y = \tan(4\theta + q)$ is:**Solution**

1. Tangent period is $\pi/|4|$.

Correct option: A**Final answer** $\pi/4$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q061 MCQ

The range of $y = 3 \sin |2x - \pi|$ is:**Solution**

1. The absolute sine is nonnegative and the factor is 3.

Correct option: B**Final answer**

$$0 \leq y \leq 3$$

Q062 Written

Express $\cos \frac{11\pi}{6}$ as a function of an acute angle.**Solution**

1. Use the QIV reference angle and cosine's positive sign.

Final answer

$$\cos \frac{\pi}{6}$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q063 Written

Simplify $\sin(\theta + \pi) - \cos(\frac{\pi}{2} - \theta)$.**Solution**

1. Replace the shifted functions by $-\sin \theta$ and $\sin \theta$.

Final answer

$$-\sin \theta - \sin \theta = -2 \sin \theta$$

Q064 Written

Simplify $\cos(\theta + \pi) + \cos(\frac{\pi}{2} - \theta)$.**Solution**

1. Use the half-turn and complementary-angle identities.

Final answer

$$-\cos \theta + \sin \theta$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q065 Written

Evaluate $\tan(-\frac{3\pi}{4})$ using an acute reference angle.**Solution**

1. The reference angle is $\pi/4$ and tangent is positive in QIII.

Final answer

1

Q066 Written

State the period and range of $y = \cos \theta$, then list its key points on one cycle.**Solution**

1. Read the standard cosine cycle at quarter-period intervals.

Final answer $T = 2\pi, -1 \leq y \leq 1; (0, 1), (\frac{\pi}{2}, 0), (\pi, -1), (\frac{3\pi}{2}, 0), (2\pi, 1)$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q067 Written

State the period and asymptotes of $y = \tan \theta$ on $[-\pi, \pi]$.**Solution**

1. Tangent is undefined at odd multiples of $\pi/2$.

Final answer

$$T = \pi; \theta = \pm \frac{\pi}{2}$$

Q068 Written

For the parent sine graph, find the x-values where $y = 0$ in $[0, 2\pi]$.**Solution**

1. Sine crosses the x-axis at integer multiples of π .

Final answer

$$0, \pi, 2\pi$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q069 Written

For the parent cosine graph, find the x-values where $y = 0$ in $[0, 2\pi]$.

Solution

1. Cosine is zero at odd quarter-turns.

Final answer

$$\frac{\pi}{2}, \frac{3\pi}{2}$$

Q070 Written

Find the period and range of $y = 4 \cos \theta$.

Solution

1. The outside factor changes the range, not the period.

Final answer

$$T = 2\pi, -4 \leq y \leq 4$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q071 **Written****Find the period and range of $y = \sin(5\theta)$.****Solution**

1. Divide the sine period by 5.

Final answer

$$T = \frac{2\pi}{5}, -1 \leq y \leq 1$$

Q072 **Written****Find the period and range of $y = \tan(\theta/4)$.****Solution**

1. Multiply the parent tangent period by 4.

Final answer

$$T = 4\pi, y \in \mathbb{R}$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q073 Written

Describe the transformations from $y = \cos \theta$ to $y = -\frac{1}{2} \cos(2\theta)$.

Solution

1. Read the inside coefficient and outside factor separately.

Final answer

horizontal compression by $1/2$, reflection in x-axis and vertical scale $1/2$; $T = \pi$

Q074 Written

Find the shift and period of $y = \sin(\theta + \frac{\pi}{6})$.

Solution

1. $\theta + p$ shifts left and does not alter period.

Final answer

left by $\frac{\pi}{6}$, $T = 2\pi$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q075 Written

Find the shift and period of $y = \cos(\theta - \frac{\pi}{2})$.**Solution**

1. θ -p shifts right by p.

Final answerright by $\frac{\pi}{2}$, $T = 2\pi$

Q076 Written

Find the central zero and period of $y = \tan(\theta + \frac{\pi}{4})$.**Solution**

1. Set the bracket equal to zero and retain tangent's period.

Final answer $\theta = -\frac{\pi}{4}$; $T = \pi$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q077 Written

For $y = 2 \sin(4\theta - \pi)$, state the shift, period and range.

Solution

- Factor 4 from the angle: $4(\theta - \pi/4)$.

Final answer

right by $\frac{\pi}{4}$, $T = \frac{\pi}{2}$, $-2 \leq y \leq 2$

Q078 Written

For $y = \cos(\theta/3 + \pi/6)$, state the shift and period.

Solution

- Rewrite the angle as $\frac{1}{3}(\theta + \pi/2)$.

Final answer

left by $\frac{\pi}{2}$, $T = 6\pi$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q079 Written

For $y = 3 \tan(2\theta + \pi)$, state the shift and period.**Solution**

1. Rewrite as $3 \tan 2(\theta + \pi/2)$.

Final answerleft by $\frac{\pi}{2}$, $T = \frac{\pi}{2}$

Q080 Written

For $y = 2 \cos(\theta/2 - \pi/8)$, state the shift, period and range.**Solution**

1. Rewrite the angle as $\frac{1}{2}(\theta - \pi/4)$.

Final answerright by $\frac{\pi}{4}$, $T = 4\pi$, $-2 \leq y \leq 2$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Quiz MCQ 1 **Concept Quiz · MCQ**

The period of $y = \sin(2\theta - \frac{\pi}{3})$ is:

Solution

1. Use $2\pi/2$.

Correct option: B

Final answer

$$\pi$$

Quiz MCQ 2 **Concept Quiz · MCQ**

$\sin(\theta + \pi)$ equals:

Solution

1. Use the half-turn identity.

Correct option: B

Final answer

$$-\sin \theta$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Quiz MCQ 3 Concept Quiz · MCQ

The range of $y = 2 \cos \theta$ is:

Solution

1. Multiply the cosine range by 2.

Correct option: B

Final answer

$$-2 \leq y \leq 2$$

Quiz MCQ 4 Concept Quiz · MCQ

The graph of $y = \cos(\theta - \frac{\pi}{4})$ is shifted:

Solution

1. $\theta - p$ shifts right by p .

Correct option: B

Final answer

right by $\frac{\pi}{4}$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Quiz WRITTEN 5 Concept Quiz · Written

For $y = 3 \sin(2\theta - \pi)$, state the horizontal shift, period and range.

Solution

1. Rewrite as $3 \sin 2(\theta - \pi/2)$.

Final answer

right by $\frac{\pi}{2}$, $T = \pi$, $-3 \leq y \leq 3$

Quiz WRITTEN 6 Concept Quiz · Written

Simplify $\sin(\theta + \frac{\pi}{2}) + \sin(\theta + \pi)$.

Solution

1. Use the quarter-turn and half-turn identities.

Final answer

$\cos \theta - \sin \theta$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Quiz WRITTEN 7 Concept Quiz · Written

Draw one cycle of $y = \tan(\theta - \frac{\pi}{3})$ and state its period and asymptotes.

Solution

1. Shift the parent tangent graph right by $\pi/3$; shift its asymptotes with it.

Final answer

$$T = \pi; \theta = \frac{5\pi}{6} + k\pi$$

Quiz WRITTEN 8 Concept Quiz · Written

For $y = -2|\sin(3x)|$, state its period and range and describe the reflection.

Solution

1. Absolute sine has input period π ; the coefficient 3 gives $\pi/3$, and -2 reflects and scales.

Final answer

$$T = \frac{\pi}{3}, -2 \leq y \leq 0; \text{ reflected below the x-axis}$$

Answer key

Use the question number shown on each page.

- Q001** (1) $-\sin \frac{\pi}{5}$ (2) 0
- Q002** (1) $\cos \frac{\pi}{6}$ (2) $-\sin \frac{\pi}{7}$ (3) $\sin^2 \theta - \cos^2 \theta$
- Q003** (1) $-\cos \frac{2\pi}{5}, -\cos \frac{2\pi}{7}, -\sin \frac{\pi}{8}, -\sin \frac{3\pi}{8}$ (2) $-\frac{1}{\sqrt{3}}, -1$ (3a) 0, (3b) $-\frac{1}{\sqrt{3}}$
- Q004** 0
- Q005** $\sin, \cos$: $T = 2\pi, -1 \leq y \leq 1$; $\tan$: $T = \pi, y \in \mathbb{R}$
- Q006** $A = 1, B = -1, C = \frac{\pi}{2}, D = 2\pi, E = \frac{1}{2}, F = \frac{5\pi}{4}$
- Q007** (1) $T_{\min} = 2\pi, T_{\max} = 2\pi, T_{\min} = \pi$; (2) read the marked extrema, intercepts and quarter-period points from the graph
- Q008** Use the standard cycles: sine/cosine have period 2π and range $[-1, 1]$; tangent has period π , range $\mathbb{R}$, and asymptotes at odd multiples of $\frac{\pi}{2}$. The requested A+F labels are read from the marked ticks and extrema in each supplied graph.
- Q009** Read x_1, x_2, x_3, x_4 from the supplied graph and add the four marked coordinates; the result is the graph-reading total.
- Q010** $y = 2 \cos \theta$: $T = 2\pi, -2 \leq y \leq 2$; $y = \tan \frac{\theta}{2}$: $T = 2\pi$
- Q011** $T = 2\pi, \frac{\pi}{2}, 4\pi$ respectively; ranges $[-2, 2], \mathbb{R}, [-1, 1]$
- Q012** (a) 2π (b) 2π (c) 2π (d) $\frac{2\pi}{3}$ (e) π (f) $\frac{\pi}{3}$ (g) 4π (h) 6π (i) 3π ; ranges follow the outside coefficient
- Q013** $T = \frac{\pi}{3}, 0 \geq y \geq -2$
- Q014** shift right by $\frac{\pi}{6}, T = 2\pi$
- Q015** Shifts: right $\frac{\pi}{3}$, left $\frac{\pi}{4}$, right $\frac{\pi}{2}$; periods $2\pi, 2\pi, \pi$.
- Q016** All sine/cosine periods are 2π ; all tangent periods are π . The shifts are read directly from the brackets.
- Q017** $y = 3 \cos(x + \frac{\pi}{3})$ since cosine is even; $T = 2\pi$
- Q018** (1) $y = 2 \cos 2(\theta - \frac{\pi}{4}), T = \pi, -2 \leq y \leq 2$; (2) $y = \tan \frac{1}{2}(\theta - \frac{\pi}{3}), T = 2\pi$
- Q019** (a) $T = 4\pi, -3 \leq y \leq 3$; (b) $T = \frac{\pi}{2}$; (c) $T = \pi, -2 \leq y \leq 2$
- Q020** Periods: (a) 4π ; (b) π ; (c) π ; (d) 2π ; (e) 2π ; (f) $\frac{\pi}{2}$; (g) π ; (h) π ; (i) 4π . Ranges are set by the outside factors.
- Q021** $y = 3|\sin(2x - \pi)|, T = \frac{\pi}{2}, 0 \leq y \leq 3$
- Q022** B
- Q023** B
- Q024** C
- Q025** A
- Q026** B
- Q027** B
- Q028** D
- Q029** D
- Q030** C
- Q031** B
- Q032** B
- Q033** B
- Q034** C
- Q035** B
- Q036** C
- Q037** B
- Q038** B
- Q039** B
- Q040** D
- Q041** A
- Q042** C
- Q043** B
- Q044** C
- Q045** B
- Q046** B
- Q047** B
- Q048** B
- Q049** B
- Q050** C
- Q051** B
- Q052** B
- Q053** A
- Q054** B
- Q055** B
- Q056** C
- Q057** B
- Q058** D
- Q059** B
- Q060** A
- Q061** B
- Q062** $\cos \frac{\pi}{6}$
- Q063** $-\sin \theta - \sin \theta = -2 \sin \theta$
- Q064** $-\cos \theta + \sin \theta$
- Q065** 1
- Q066** $T = 2\pi, -1 \leq y \leq 1$; (0, 1), $(\frac{\pi}{2}, 0)$, $(\pi, -1)$, $(\frac{3\pi}{2}, 0)$, $(2\pi, 1)$
- Q067** $T = \pi; \theta = \pm \frac{\pi}{2}$
- Q068** 0, $\pi, 2\pi$
- Q069** $\frac{\pi}{2}, \frac{3\pi}{2}$
- Q070** $T = 2\pi, -4 \leq y \leq 4$
- Q071** $T = \frac{2\pi}{5}, -1 \leq y \leq 1$
- Q072** $T = 4\pi, y \in \mathbb{R}$
- Q073** horizontal compression by 1/2, reflection in x-axis and vertical scale 1/2; $T = \pi$
- Q074** left by $\frac{\pi}{6}, T = 2\pi$
- Q075** right by $\frac{\pi}{2}, T = 2\pi$
- Q076** $\theta = -\frac{\pi}{4}; T = \pi$
- Q077** right by $\frac{\pi}{4}, T = \frac{\pi}{2}, -2 \leq y \leq 2$
- Q078** left by $\frac{\pi}{2}, T = 6\pi$
- Q079** left by $\frac{\pi}{2}, T = \frac{\pi}{2}$
- Q080** right by $\frac{\pi}{4}, T = 4\pi, -2 \leq y \leq 2$
- Quiz MCQ 1** B
- Quiz MCQ 2** B
- Quiz MCQ 3** B

Quiz MCQ 4 B**Quiz WRITTEN 6** $\cos \theta - \sin \theta$ **Quiz WRITTEN 7** $T = \pi; \theta = \frac{5\pi}{6} + k\pi$ **Quiz WRITTEN 8** $T = \frac{\pi}{3}, -2 \leq y \leq 0$; reflected below the**Quiz WRITTEN 5** right by $\frac{\pi}{2}$, $T = \pi, -3 \leq y \leq 3$